When paperwork agrees, physical reality can disagree.

A GST officer can’t tell which factories are hiding production or inflating tax credits, because every document she checks is declared by the factory itself, while the independent evidence (power use, goods movement, supplier records, bank flows) sits in separate places and is never compared. If this worked, she would start each week with a short list of units whose paperwork contradicts physical reality, each contradiction spelled out, and honest-but-unusual factories cleared.

Decision support for human review

Impact and the current gap

This prototype targets officers reviewing declarations when supporting records arrive separately.

What exists today

Invoice matching compares tax documents. We add operational records; no benchmark of commercial fraud systems is claimed.

What this project adds

It puts production, movement, supplier and bank evidence on one factory-month record, with the evidence rows visible to the reviewer.

Why it matters

Prioritise specific contradictions and preserve an explanation for honest operational changes.

Method and the two-signal rule

Six SQL checks form four source families. Displayed notes use a validated deterministic fallback; an LLM is optional.

01

Production

Material, power and stock

02

Movement

Dispatch, invoice and reuse

03

Supplier

Supplier returns; loop context

04

Financial

Bank vs sales and purchases

PRIORITISE ONLY WHEN AT LEAST TWO SOURCE FAMILIES FIRE

A single-source month stays review-only or receives a documented cleared disposition.

V3 held-out results and limits

V3-A selected threshold 0.05 and K=3. V3-C confirms frozen choices.

V3-C 95% CIs: recall 78.88% – 83.64%; FPR 2.06% – 2.49%.
V3-B primary: recall 81.14%; FPR 2.20%.
Wrong-clears: 0 of 720 active-cheat factories; approximate upper bound 0.42%. Assumes independent factories.

81.24%
Factory-month recall

2.28%
Factory-month false-positive rate

97.52%
Factory queue precision, K=3

605 factories enter the queue. Top-50 monthly macro precision: 75.00%.

Synthetic only. Verification labels are generated. Forger test, ablation intervals and second-machine reproduction remain open. Human review required.

Exasol Personal and the data-access roadmap

The database runs the evidence views locally. The proposal starts with tax-system data and adds other sources only under lawful access.

172,800 current V3-C feature rows

2,400 factories. Stock median: Exasol 0.169 s; DuckDB 0.006 s.
DuckDB was faster. MacBook workload timings.

Data-access roadmap

1. Tax-system data
Invoices, e-way bills, supplier returns

2. Lawful additional evidence
Power and bank records under authorised access and privacy controls

The 20.98M-row run is historical DEV, not fresh. Current timings exclude Exasol ingestion; DuckDB load is separate. No German deployment is claimed.